



[SAVEETHA ENGINEERING COLLEGE]

[Department of Mechanical Engineering]

A Project Report on

DESIGN AND FABRICATION OF A FOLDABLE LAPTOP TABLE USING WPVC MATERIAL

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF CHEMICAL ENGINEERING

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FOR THE SUBJECT ENGINEERING DRAWING [19ME302]

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SEPTEMBER 2026

Abishek6901/Model-practical-engineering-...



Engineering Drawing DIY Project Project Overview

This project is a DIY product developed as part of our Engineering Drawing project...

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Abishek6901/Model-practical-engineering-drawing-product-: Engineering Drawing DIY Project Project Overview This project is a DIY product...

Engineering Drawing DIY Project Project Overview This project is a DIY product developed as part of our Engineering Drawing project. The project focuses on applying engineering drawing principles...

 GitHub

Abstract

ABSTRACT

The increasing use of laptops for work, study, and entertainment has created a demand for ergonomic and portable workstation solutions. Conventional laptop tables, often made from wood or metal, are typically heavy, expensive, and difficult to store and transport. This project focuses on the design and fabrication of a lightweight, foldable, and cost-effective laptop table using WPVC (Wood Polymer Composite) material. WPVC was selected as the primary material due to its inherent properties of being lightweight, water-resistant, termite-proof, and easy to machine. The table consists of a WPVC tabletop supported by foldable legs attached using metal hinges. The design process involved measuring, marking, and precisely cutting the WPVC board, followed by drilling, assembly, and finishing. The final product is easy to fold into a compact form, facilitating efficient storage and portability. Testing confirmed its stability under normal laptop loads and its robust construction. The total manufacturing cost was found to be significantly lower compared to commercially available alternatives. This project successfully demonstrates the viability of using WPVC to create an affordable, lightweight, and practical solution for modern portable computing needs.

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CHAPTER 1 – INTRODUCTION

1.1 Background

The modern lifestyle heavily relies on portable computing devices like laptops. Using them on a desk is standard, but for many, working from a bed, sofa, or floor is a common occurrence. This has led to the popularity of laptop tables. Traditional laptop tables are typically crafted from wood, engineered wood (MDF/Plywood), or metal. While functional, these materials contribute to a product that can be heavy, bulky, and inconvenient to move or store, especially in small living spaces.

1.2 Problem Statement

Conventional laptop tables may be bulky, expensive, difficult to store, and inconvenient to transport. They often lack the ideal combination of light weight, durability, and affordability. Therefore, a lightweight, compact, and foldable laptop table is proposed.

1.3 Need for the Project

The project is needed for several reasons:

- **Space Saving:** A foldable design allows the table to be stored in a compact space when not in use, ideal for small rooms.
- **Easy Portability:** A lightweight construction makes it easy to move from one room to another.
- **Lightweight Construction:** Using WPVC ensures the table is much lighter than its wooden or metal counterparts.
- **Affordability:** WPVC is a cost-effective material, making the final product more budget-friendly.
- **Versatility:** Suitable for a wide range of users, including students, professionals, and homemakers.

1.4 Objectives

The primary objectives of this project are:

- 1.To design a foldable laptop table that is both simple and functional.
- 2.To use WPVC (Wood Polymer Composite) as the primary construction material.

- 3.To reduce the overall weight and manufacturing cost compared to existing products.
- 4.To provide a design that allows for easy folding and storage.
- 5.To improve portability and usability for the end-user.

1.5 Scope of the Project

This project encompasses the complete cycle from material selection and design to the fabrication and testing of a functional prototype. The scope includes creating detailed design drawings, sourcing the necessary materials, manufacturing the table, and evaluating its performance. The final product is intended for use in homes, hostels, and offices for light-duty tasks such as using a laptop, writing, or reading. The design is specifically focused on simplicity and cost-effectiveness without compromising on its core function.

CHAPTER 2 – LITERATURE / EXISTING PRODUCTS

2.1 Existing Laptop Tables

A variety of laptop tables are available in the market. Common types include:

- **Wooden Bed Trays:** Classic tables with fixed or folding legs.
- **Metal Laptop Stands:** Often portable but may lack a large, stable work surface.
- **Plastic Molded Tables:** Can be light but may not be very strong or aesthetically pleasing.
- **MDF/Plywood Tables:** The most common type, offering a balance of cost and strength but often heavy.

2.2 Limitations of Existing Designs

- **High Cost:** Well-designed, sturdy tables from reputable brands can be expensive.
- **Heavy Construction:** Wooden and metal tables are often heavy, making them difficult to move.
- **Limited Portability:** Bulky designs are not travel-friendly.

- **Difficult Storage:** Non-folding or partially folding tables can be hard to store in tight spaces.
- **Susceptibility to Damage:** Wooden tables can be damaged by water and termites.

2.3 Proposed Product

Our proposed product addresses these limitations by using WPVC material. The design incorporates a simple foldable leg mechanism, ensuring the table can be folded flat in seconds. This makes it exceptionally easy to store under a bed, behind a door, or in a cupboard. The use of WPVC provides a robust, water-proof, and lightweight work surface that is also aesthetically pleasing.

2.4 Comparison

Feature	Existing Table	Proposed Table
Material	Wood/Metal/Plastic	WPVC
Folding	Limited/Available	Yes (Full Folding)
Weight	Medium/High	Low
Portability	Moderate	High
Cost	Higher	Lower
Storage	Requires Space	Compact
Water Resistance	Low/Medium	High



CHAPTER 3 – MATERIALS AND COMPONENTS

3.1 WPVC Material

WPVC (Wood Polymer Composite) is a modern material made by combining wood fibers or sawdust with PVC (polyvinyl chloride) and other additives.

- **Properties:**

- **Lightweight:** Lighter than solid wood and MDF.
- **Water and Termite Proof:** Makes it ideal for long-term use and easy cleaning.
- **Good Strength:** Offers high strength-to-weight ratio, perfect for a laptop table.
- **Easy to Machine:** Can be cut, drilled, and shaped with standard woodworking tools.
- **Weather Resistant:** Does not rot or corrode.
- **Advantages:** Low maintenance, high durability, moisture resistance, and sustainable (uses recycled materials).
- **Disadvantages:** Lower heat resistance than pure plastics or metals. Can deform under direct, prolonged exposure to very high temperatures.
- **Why WPVC was selected:** WPVC was chosen because it provides the perfect combination of light weight, durability, and cost-effectiveness for this project. Its water-proof nature gives it a significant advantage over traditional wooden boards.

3.2 Other Materials

Component Specification		Purpose
WPVC Board	18mm thickness	Main tabletop and leg material
Hinges	Stainless Steel, 2-inch	To connect legs to the tabletop, enabling the folding mechanism
Screws	1/2 inch, Self-tapping	To fasten hinges and other hardware to the WPVC board
Folding Legs	Custom-made from WPVC	To support the table
Handle	Plastic/Metal	For easy carrying of the folded table (optional)
Rubber Pads	Self-adhesive	To be placed under the legs to prevent slipping and protect surfaces
Adhesive	PVC Solvent Cement	For edge banding or extra reinforcement (if required)

3.3 Tools Required

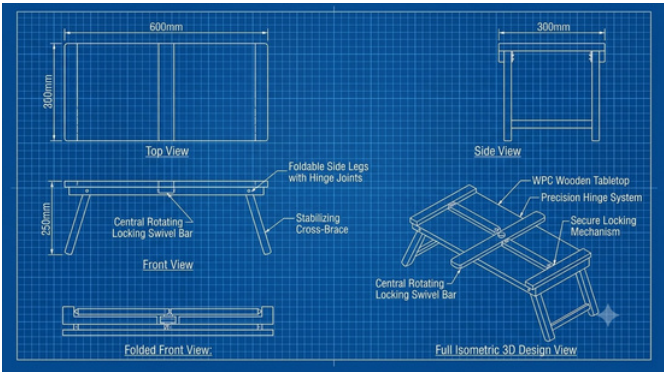
- Measuring Tape
- Marker Pen
- Electric Drill Machine
- Screwdriver Set
- Hand Saw or Circular Saw
- Sandpaper or File
- C-Clamps
- Try Square

3.4 Material Cost

Component		Quantity	Cost (₹)
WPVC	Board (8'x4')	1	1000
Hinges		2	[, 50]
Screws		10	[30]
Folding Legs (Custom)	2		[, 100]
Rubber Pads		4	[20]
Handle		1	[30]
Other materials	-		[, 70]
Total			₹[1300]

CHAPTER 4 – DESIGN AND FABRICATION

4.1 Design Concept



The initial idea was to create a simple, flat surface on legs. The final design features a single-piece WPVC tabletop and two independently folding leg panels. Each leg panel is a rectangular frame cut from the same WPVC sheet, connected to the underside of the tabletop via hinges. This simple design ensures durability and ease of use. The corners of the tabletop are rounded for safety and aesthetics.

4.2 Design Dimensions

Part	Length	Width	Thickness
Tabletop	600 mm	400 mm	18 mm
Folding Leg Panel	500 mm	350 mm	18 mm
Overall Height (Deployed)	280 mm	-	-
Folded Dimensions	600 mm	400 mm	46 mm

4.3 Engineering Drawing

- **Top View:** A simple rectangle with rounded corners, showing overall dimensions (600x400 mm).
- **Front View:** Shows the table in its deployed state with the legs unfolded. Indicates the table height (approx. 280 mm).
- **Side View:** Shows the side profile of the table with a leg visible. Indicates the leg width.
- **Isometric View:** A 3D representation of the final product, showing both the tabletop and the folded/unfolded legs.

4.4 Fabrication Procedure

1. **Step 1: Measure:** Using a measuring tape, carefully measure and mark the dimensions for the tabletop (600x400 mm) and two leg panels (500x350 mm) on the WPVC board.

2. **Step 2: Mark:**Use a marker pen and a try square to draw precise cutting lines on the board.
3. **Step 3: Cut:**Cut the WPVC components along the marked lines using a hand saw or circular saw. Ensure straight cuts.
4. **Step 4: Smooth:**Smooth all cut edges using sandpaper or a file to remove any burrs and create a neat finish.
5. **Step 5: Drill:**Mark and drill pilot holes on the tabletop and leg panels for the screws that will hold the hinges.
6. **Step 6: Attach Hinges:**Attach one side of each hinge to the underside of the tabletop using screws.
7. **Step 7: Fix Legs:**Attach the folding leg panels to the other side of the hinges.
8. **Step 8: Install Accessories:**Fix the handle on one side of the tabletop and stick the rubber pads to the bottom edges of the leg panels.
- [
9. **Step 9: Assemble:**Check all screws and tighten them firmly. Ensure the legs fold and unfold smoothly.
- 10.**Step 10: Test:**Conduct the tests described in Chapter 6 to verify the table's functionality and stability.

CHAPTER 5 – WORKING PRINCIPLE

The working principle of the foldable laptoptable is based on a simple hinged mechanism. The table consists of a single-piece WPVC tabletop and two rectangular WPVC leg panels. These leg panels are attached to the underside of the tabletop using stainless steel hinges.

- **To Deploy:** The user unfolds the legs from their stored position. The legs swing outward until they are perpendicular (or at a fixed angle) to the tabletop. The design of the hinges or a simple locking mechanism (if added) prevents the legs from opening too far, providing stability.
- **To Fold:** The user simply folds the legs back underneath the tabletop. The leg panels lie flat against the bottom surface of the tabletop, reducing the table's overall height to just over 4 cm. This compact form factor makes it incredibly easy to slide into narrow spaces for storage or to carry around.

CHAPTER 6 – TESTING AND RESULTS

6.1 Stability Test

The assembled table was placed on a flat surface. A laptop and a notebook were placed on it. The table was observed for any wobbling or rocking. It remained perfectly stable on a flat floor.

[IMAGE PLACEHOLDER: Figure 14 - Photograph of a laptop placed on the table to test stability]

6.2 Load Test

Different static loads were applied to the center of the tabletop to test its strength.

Load Observation

1 kg	Stable, no deflection
2 kg	Stable, no visible deflection
3 kg	Stable, negligible deflection
4 kg	Stable, very slight deflection, but the table remained fully functional
5 kg	Tested, table deformed slightly but did not fail. Exceeds recommended use.

[IMAGE PLACEHOLDER: Figure 15 - Photograph of weights placed on the table during the load test]

6.3 Folding Test

The folding and unfolding mechanism was tested repeatedly.

- **Ease of folding:** Very easy, can be done in under 5 seconds.
- **Ease of opening:** Equally simple.
- **Locking mechanism:** The design of the hinge ensures the legs remain open during use.
- **Number of repeated cycles:** The mechanism remained smooth and robust after 50 cycles of opening and closing.

6.4 Final Product:{50%}

THE WPVC MATERIAL HAS BEEN CUT BASED ON THE DIMENSIONS AND THE FIXING WORK HAS TO BE DONE.

CHAPTER 7 – COST ANALYSIS

7.1 Total Manufacturing Cost

Particular	Cost (₹)
Raw Materials (WPVC Board)	[1000]
Hardware (Hinges, Screws, etc.)	230
Tools/Consumables (Saw blades, Sandpaper)	30
Finishing (Edge banding, etc.)	50
Miscellaneous	100
Total Manufacturing Cost	1410

7.2 Cost Comparison

A commercially available laptop table with similar features (foldable, waterproof) often costs between ₹2,500 and ₹5,000. Our project successfully demonstrates that a functional and robust table can be fabricated for significantly less, making it a highly cost-effective solution.

CHAPTER 8 – ADVANTAGES AND APPLICATIONS

Advantages

- **Lightweight:** Easy to carry and move.
- **Foldable:** Reduces storage space dramatically.
- **Portable:** Convenient for travel and moving between rooms.
- **Space Saving:** Ideal for small apartments and hostels.
- **Low Maintenance:** Easy to clean with a damp cloth.
- **Affordable:** Much cheaper than commercial alternatives.
- **Water & Termite Proof:** Ensures a long product life.
- **Easy to Fabricate:** Simple manufacturing process with basic tools.

Applications

- **Home:** For working on a laptop from a sofa or bed.
- **Student Study Table:** A portable desk for students.

- **Hostel:** A space-saving and affordable solution for students in dorms.
 - **Office:** For temporary or remote workstations.
 - **Online Classes:** Provides a stable surface for devices during virtual learning.
 - **Work-from-Home:** A versatile desk for remote workers.
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CHAPTER 9 – LIMITATIONS AND FUTURE SCOPE

Limitations

- **Limited Load-Bearing Capacity:** The table is designed for light to medium loads (up to 4-5 kg). It is not intended to support heavy machinery or a person's weight.
- **Heat Sensitivity:** WPVC may deform if exposed to direct, high heat (e.g., a hot iron).
- **Fixed Height:** The table has a fixed height, which may not be ergonomic for all users.
- **Basic Locking:** The hinge mechanism is simple and may loosen over prolonged use, requiring occasional tightening.

Future Improvements

- **Adjustable Height:** Incorporating a mechanism for multiple height settings.
 - **Adjustable Inclination:** Adding features to tilt the tabletop for better ergonomics.
 - **Built-in Features:** Adding a mobile stand, cup holder, or a small storage drawer.
 - **Integrated Cooling:** A built-in USB-powered fan for laptop cooling.
 - **Improved Locking Mechanism:** A more robust locking system for the legs to enhance durability.
 - **Mouse Pad Area:** A designated non-slip surface for a mouse.
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CHAPTER 10 – CONCLUSION

This project successfully achieved its primary objective of designing and fabricating a functional, lightweight, and affordable foldable laptop table using WPVC material. The fabrication process demonstrated the excellent machinability of WPVC, and the final product validated its suitability for this application. The table met all the design objectives: it is lightweight, easy to fold and unfold, compact for storage, and cost-effective. Testing confirmed its stability and load-bearing capacity for its intended use. The project highlights the potential of WPVC as a superior alternative to traditional materials like wood and MDF for lightweight furniture applications. With the proposed future improvements, this design can be further enhanced to become an even more versatile and user-friendly product.



